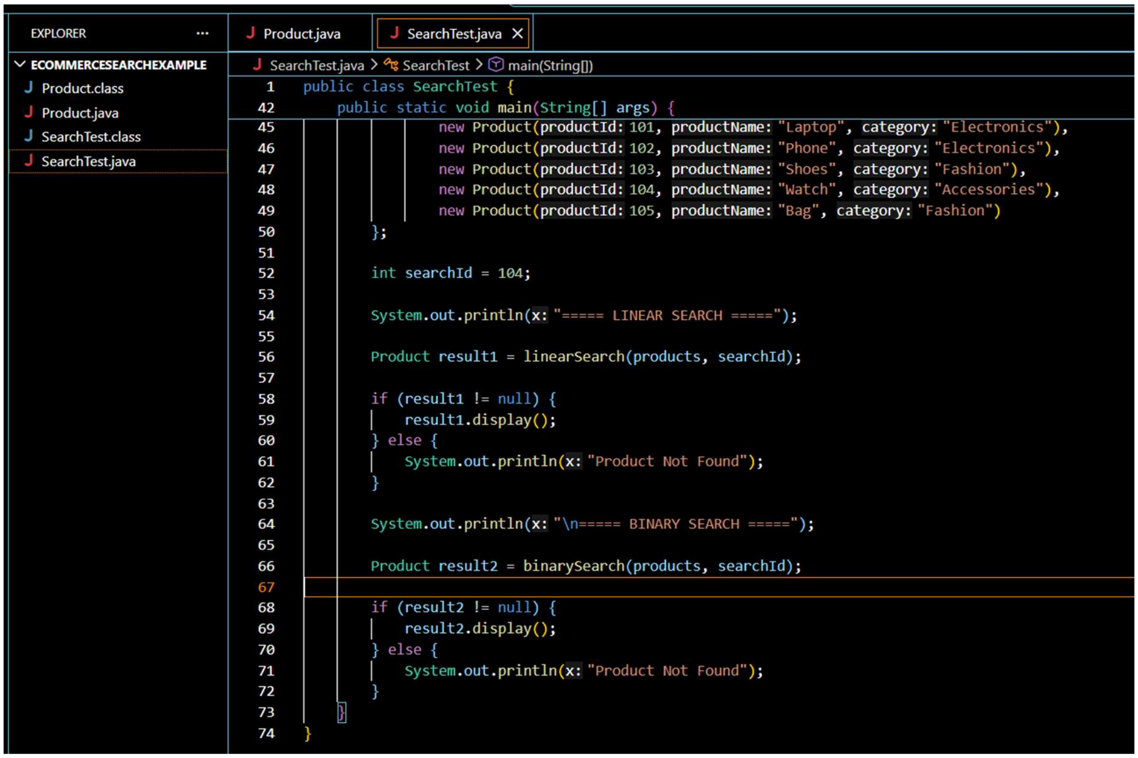
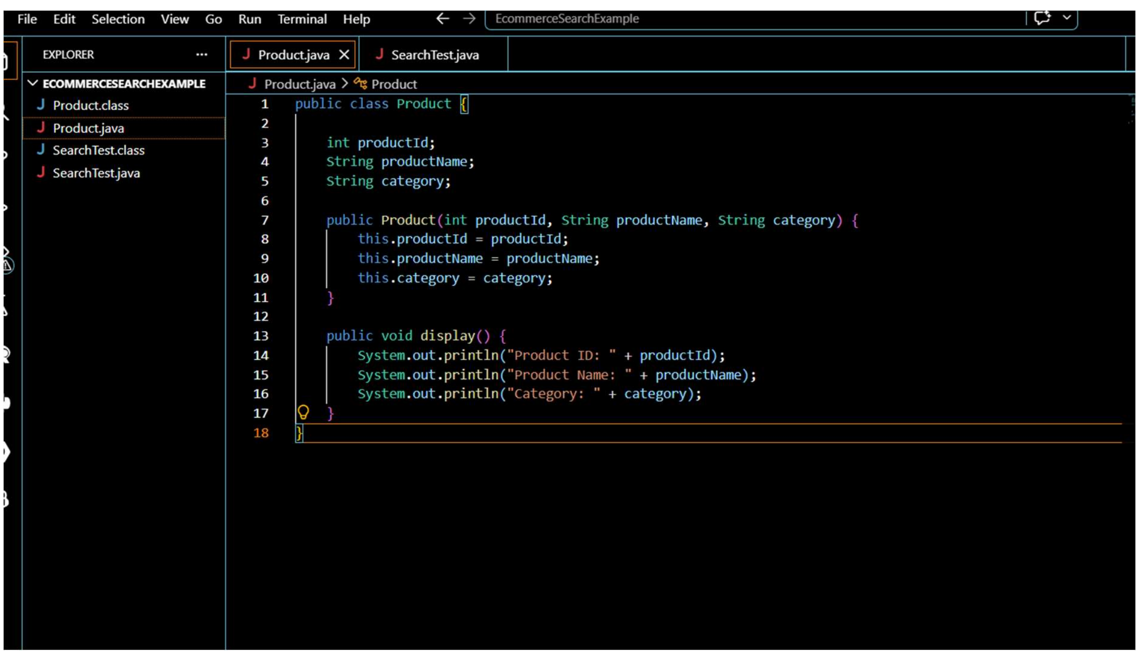
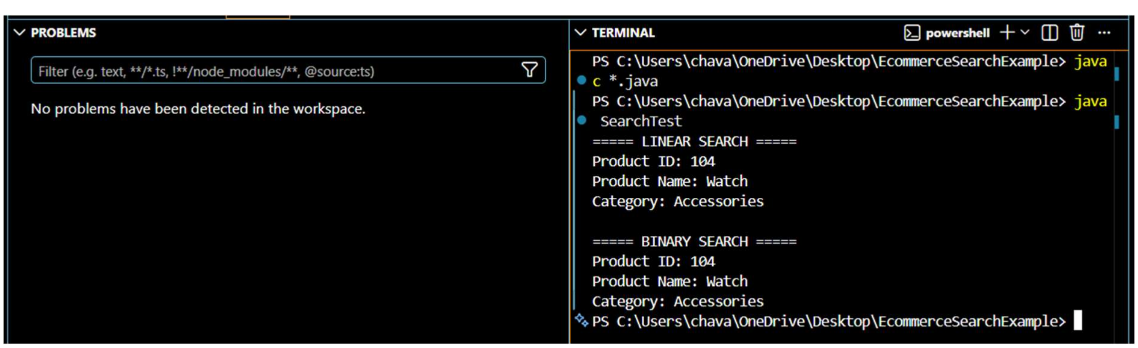
Exercise 2: E-commerce Platform Search Function







Exercise 7: Financial Forecasting

```
EXPLORER ...
FINANCIALFORECASTING
  ForecastTest.class
  ForecastTest.java
TIMELINE
```

```
Welcome    ForecastTest.java
ForecastTest.java > ForecastTest
```

```java
public class ForecastTest {

    // Recursive method
    public static double forecast(double currentValue,
                                  double growthRate,
                                  int years) {

        // Base case
        if (years == 0) {
            return currentValue;
        }

        // Recursive case
        return forecast(
                currentValue * (1 + growthRate),
                growthRate,
                years - 1);
    }

    Run | Debug
    public static void main(String[] args) {

        double currentValue = 10000;
        double growthRate = 0.10; // 10%
        int years = 5;

        double futureValue =
                forecast(currentValue,
                         growthRate,
                         years);

        System.out.println("Current Value = ₹" + currentValue);
        System.out.println("Growth Rate = " + (growthRate * 100) + "%");
        System.out.println("Years = " + years);
        System.out.println("Predicted Future Value = ₹" + futureValue);
    }
}
```

TERMINAL   powershell

```
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting> javac ForecastTest.java
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting> java ForecastTest
Current Value = â??10000.0
Growth Rate = 10.0%
Years = 5
Predicted Future Value = â??16105.100000000008
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting>
```